

Pearson vs. Distance Correlation for ABCD Background Estimation

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We are testing whether the two NN discriminants used in the ABCD background estimate behave as statistically independent variables. The natural first instinct is to report a Pearson correlation; the measured value is $\rho \approx 0.3$. This note argues that Pearson is the wrong statistic for the question being asked, that distance correlation (dCor) is the right one, and that the failure of Pearson here is not incidental but *structural*: the properties of NN discriminants are precisely those that void Pearson's guarantees. Three parts follow — (1) what the two metrics measure and how they differ, (2) the dependence each can detect that the other cannot, and (3) why this experiment is inherently a dCor problem.

One line for the meeting: Pearson can only *fail to find* dependence; dCor can *certify its absence*. ABCD's null hypothesis is independence, so we need the statistic that certifies absence.

1 The two metrics and their differences

Pearson — linear, signed, Gaussian-bound

The Pearson coefficient is the normalised covariance of the raw values,

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}. \quad (1)$$

Geometrically it is the cosine of the angle between the two mean-centred data vectors — a single inner product. It therefore answers exactly one question: *when X rises above its mean, does Y rise proportionally, along a straight line?* It is cheap, bounded in $[-1, 1]$, and *signed* (it reports the direction of the trend). Its defining limitation is the direction of its guarantee:

$$X \perp Y \implies \rho = 0, \quad \text{but} \quad \rho = 0 \not\Rightarrow X \perp Y. \quad (2)$$

The converse holds *only* under joint Gaussianity. Outside the Gaussian world, $\rho = 0$ carries no independence content.

Distance correlation — nonlinear, unsigned, distribution-free

Distance correlation discards the raw values and works on *pairwise distances*. For samples $\{X_j\}$, form the distance matrix $a_{jk} = |X_j - X_k|$ and double-centre it,

$$A_{jk} = a_{jk} - \bar{a}_{j\cdot} - \bar{a}_{\cdot k} + \bar{a}_{\cdot\cdot}, \quad (3)$$

and likewise B_{jk} from $\{Y_j\}$. The (squared) distance covariance and the distance correlation are

$$\text{dCov}^2(X, Y) = \frac{1}{n^2} \sum_{j,k} A_{jk} B_{jk}, \quad \text{dCor}(X, Y) = \frac{\text{dCov}(X, Y)}{\sqrt{\text{dCov}(X, X) \text{dCov}(Y, Y)}} \in [0, 1]. \quad (4)$$

The question it asks is geometric: *when two events are far apart in X , are they also far apart in Y ?* That is the definition of dependence itself, and it is blind to whether the relationship is a line, a curve, a fork, or a ring. The double-centring is the load-bearing step — it is what makes the key property hold for *any* distribution:

$$\text{dCor}(X, Y) = 0 \iff X \perp Y. \quad (5)$$

Side by side

	Pearson ρ	Distance correlation
Built from	raw values (covariance)	pairwise distances
Detects	linear association only	any dependence
Range	$[-1, 1]$, <i>signed</i>	$[0, 1]$, magnitude only
Zero means	no <i>linear</i> trend	<i>independence</i> (always)
Independence test	valid only if jointly Gaussian	valid for any distribution
Cost	$O(n)$	$O(n^2)$ (subsample for large n)

A useful unifying fact: *under* joint Gaussianity, dCor is a deterministic monotonic function of $|\rho|$ — the two carry the same information in that regime. They disagree precisely to the extent the data departs from Gaussian. The gap between them is itself a non-Gaussianity meter.

2 What each can catch that the other cannot

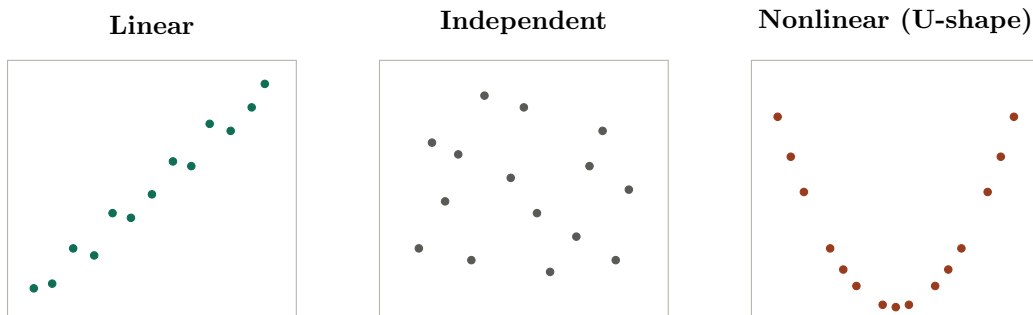
The detectable-dependence relation *nests*:

$$\rho \neq 0 \implies \text{dCor} > 0, \quad \text{dCor} = 0 \implies \rho = 0. \quad (6)$$

Everything Pearson can flag, dCor also flags — plus an entire class (nonlinear and higher-moment dependence) that Pearson cannot. As an independence detector, Pearson’s coverage is a strict subset of dCor’s. The one thing dCor lacks is the *sign*: it reports how much dependence, never which direction. For a factorisation test that asymmetry is exactly the right trade — we care whether dependence exists, not its sign.

The canonical miss: zero ρ , total dependence

Let X be symmetric about its mean and $Y = (X - \frac{1}{2})^2$. Then Y is *completely determined* by X , yet the left and right halves contribute opposite-sign covariance and cancel: $\rho = 0$ exactly. dCor is large. Pearson would issue a clean bill of health on a pair that ABCD cannot close. The figure below shows the three regimes that matter.



	Linear	Independent	Nonlinear (U)
Pearson ρ	0.95	≈ 0.02	≈ 0.00
dCor	0.95	≈ 0.04	0.52
Verdict	agree	agree	diverge — Pearson blind

The two metrics agree when the relationship is linear (left) and agree when the variables are independent (centre). They diverge only on nonlinear dependence (right) — and that is exactly the regime our discriminants live in.

The silent miss: variance dependence

Even if the conditional *mean* $\mathbb{E}[Y | X]$ is flat in X (so $\rho \approx 0$), the conditional *spread* $\text{Var}(Y | X)$ can depend strongly on X . NN confidence is heteroscedastic across input space almost always. Pearson is blind to second-moment dependence; dCor catches it, because changing spread changes pairwise distances. ABCD assumes the *full* conditional factorises, not merely the conditional mean — so variance dependence breaks closure while Pearson reports zero. This is the most easily overlooked failure mode and is worth probing explicitly with a sliced scan.

3 Why our experiment is inherently dCor

The ABCD null is an independence statement

The ABCD prediction $A_{\text{pred}} = BC/D$ is exact if and only if the background joint factorises in the two discriminants, $P(x, y) = P(x)P(y)$. That is the definition of independence. So “does ABCD close?” and “are the discriminants independent in the background?” are the same question — and only dCor answers it with a guarantee. Pearson can return a comforting 0 on a background that will not close; dCor cannot.

The information-theoretic identity (HELIX / MaxEnt view)

The factorised joint $P(x)P(y)$ is exactly the maximum-entropy distribution consistent with the two marginals: if all we constrain are $P(x)$ and $P(y)$, Jaynes’ principle returns the product as the least-committal joint. The deviation from that null is the mutual information, and it is an identity, not an analogy:

$$I(X; Y) = D_{\text{KL}}(P(x, y) \| P(x)P(y)), \quad I(X; Y) = 0 \iff X \perp Y. \quad (7)$$

Mutual information *is* the distance from the MaxEnt null; dCor is its well-behaved, estimator-friendly cousin sharing the iff-independence property. This is why Pearson is disqualified at the level of principle, not merely performance: it does not measure distance from the factorised null, so it cannot be the correct statistic even in principle. If a correlation constraint is added to the marginals, MaxEnt returns a single exponential tilt $P(x, y) \propto P(x)P(y)e^{\lambda g(x, y)}$, with $\lambda = 0$ recovering closure — which turns any residual non-closure into a measurable parameter (future work; the proof target).

Why NN1 and NN2 leave the Gaussian box

Pearson’s independence guarantee requires joint Gaussianity. NN discriminants violate that assumption through several independent channels at once:

- **Shared-latent nonlinearity.** Both networks read the same event (same displaced MS vertex, same kinematics). Two different nonlinear functions of a common cause generically produce nonlinear, often non-monotone dependence — the curvature Pearson cannot see.

- **Output squashing.** A classifier score lives in $[0, 1]$ and piles up against the rails; the joint clusters in the corners of the plane, far from a Gaussian ellipse. A sigmoid output is the first thing to leave the Gaussian box.
- **Tail co-movement.** The ABCD signal region is a high-score *corner* — a tail, not the bulk. Variables can be linearly uncorrelated overall yet co-move in the tails. Pearson is bulk-weighted and dilutes exactly the region that decides closure.
- **Heteroscedasticity.** Confidence spread varies across input space; second-moment dependence is invisible to ρ (see §2).
- **Barrel quantization.** Integer-valued inputs place the score on a discrete lattice — maximally non-Gaussian, with flat-then-jump structure at the cut boundaries. This is the signature of the barrel 1.79 non-closure. (Caveat: quantization also biases dCor/MI estimation, so evaluate on the *continuous* pre-quantization scores.)

Each bullet is a way of being non-Gaussian, and non-Gaussian is exactly where $\rho = 0$ stops meaning independent. The discriminants violate the assumption through *several* channels simultaneously — so it is not that Pearson *might* mislead here; the setting is built to make it mislead. dCor evaluates them correctly because its zero is an independence certificate regardless of distributional shape.

Practical recipe

On *background-only, continuous* scores: (i) global dCor with a permutation test for significance; (ii) sliced dCor across NN1 to localise where the dependence lives; (iii) the bootstrapped ABCD closure ratio as the physics arbiter. Pearson may be reported alongside, but only as a non-Gaussianity cross-check — never as the basis for the independence decision.

Honest qualifier: the arguments above establish why the independence assumption cannot be made for these variables, not that the dependence is necessarily large. Which channel dominates — tail, variance, or quantization — is what the sliced dCor scan and conditional-shape check are there to determine.